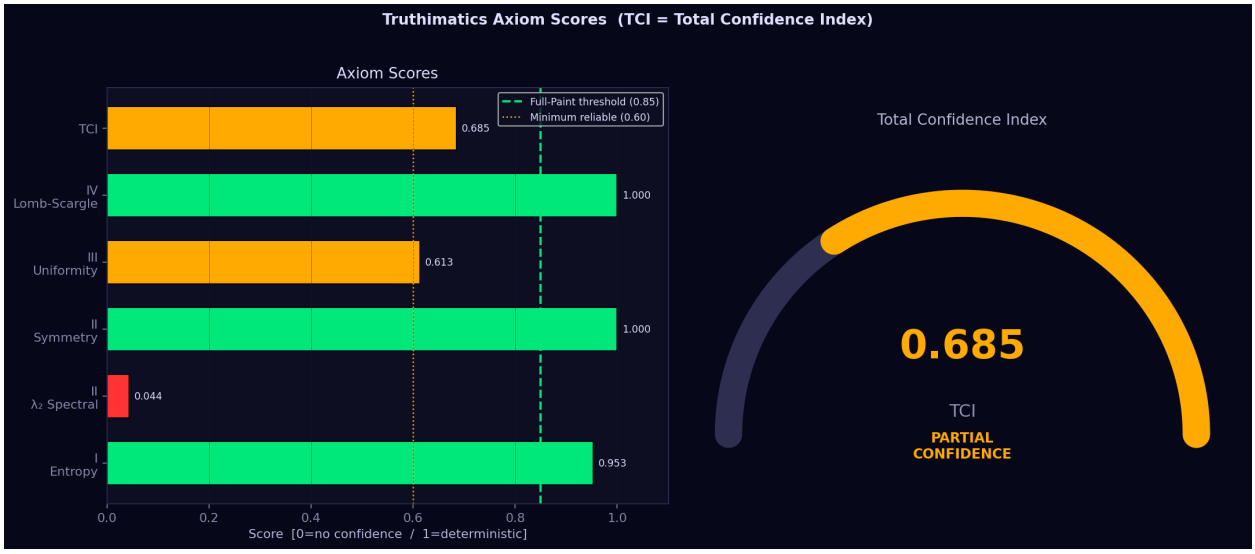
# AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

| | |
|---|---|
| **Specimen ID** | AXIOM_B3A0022558 |
| **Taxa** | Ceratosaurus |
| **Clade** | Theropod |
| **PBDB Records** | 29 |
| **Mass (regression)** | 2288.6 kg (Campione & Evans 2012) |
| **Mass (volumetric)** | 2108.1 kg (Silhouette integral, $\rho$=1060.0 kg/m$^3$) |
| **Body Volume** | 2.480 m$^3$ |
| **Validity Score** | 1.0000 / 1.0000 |
| **TCI** | 0.6848 (partial) |
| **Axiom I Entropy** | 0.9531 |
| **Axiom II $\lambda$■** | 0.1763 (symmetry 100%) |
| **Axiom III Uniformity** | 0.6133 |
| **Axiom IV Lomb FAP** | 1.000000 peak 0.000 Hz |
| **Integrity** | 56.0% Measured / 44.0% Inferred |
| **Elements** | 25 |
| **CoG (m)** | [0.328, 0.522, 0.0] |
| **Generated** | 2026-04-30 23:52 UTC |

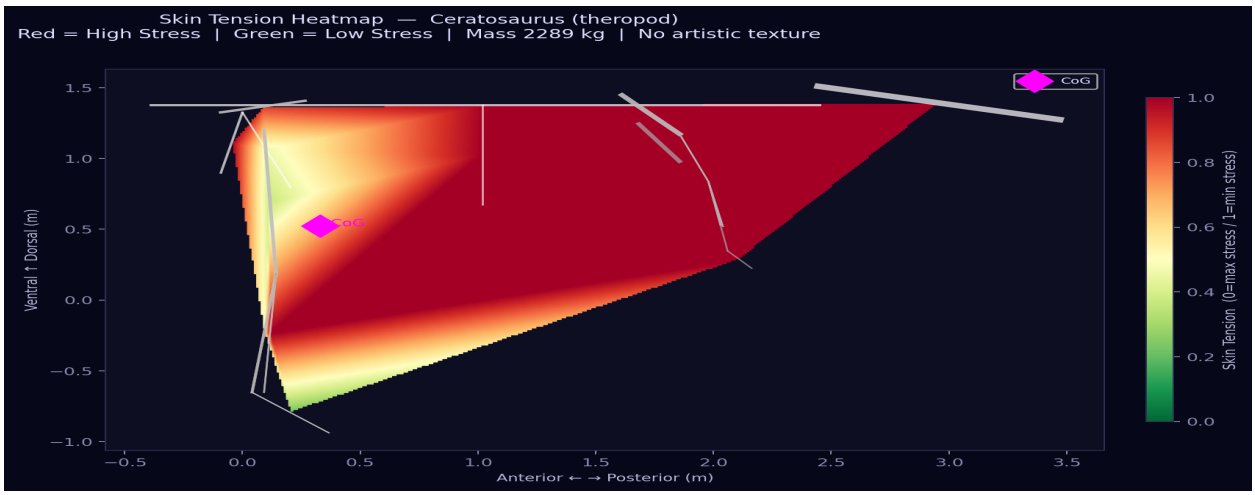
# 1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



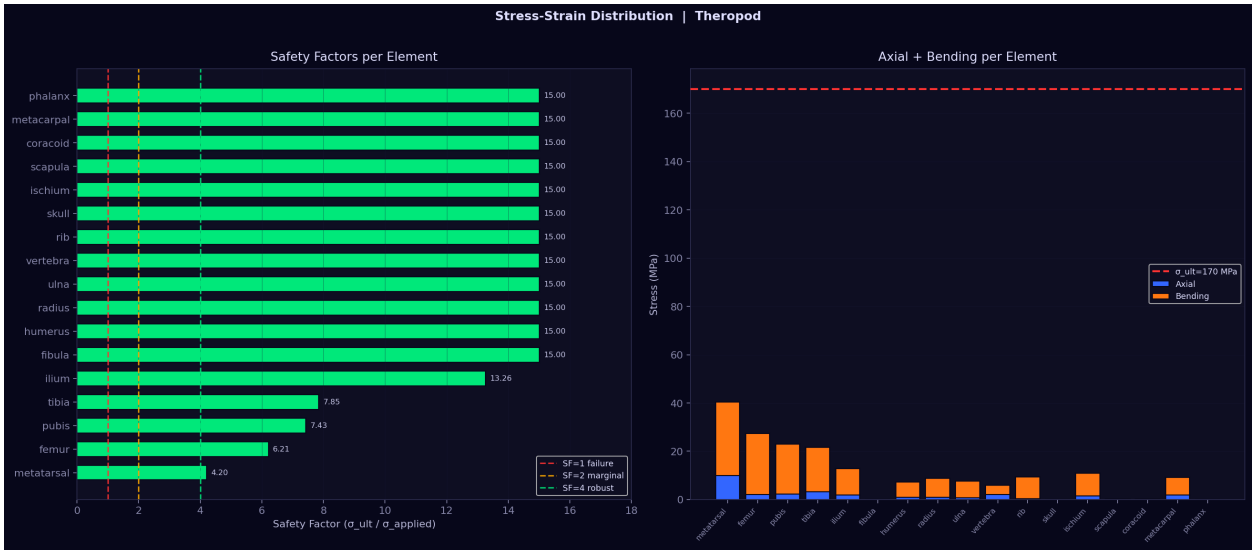
## 2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



# 3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

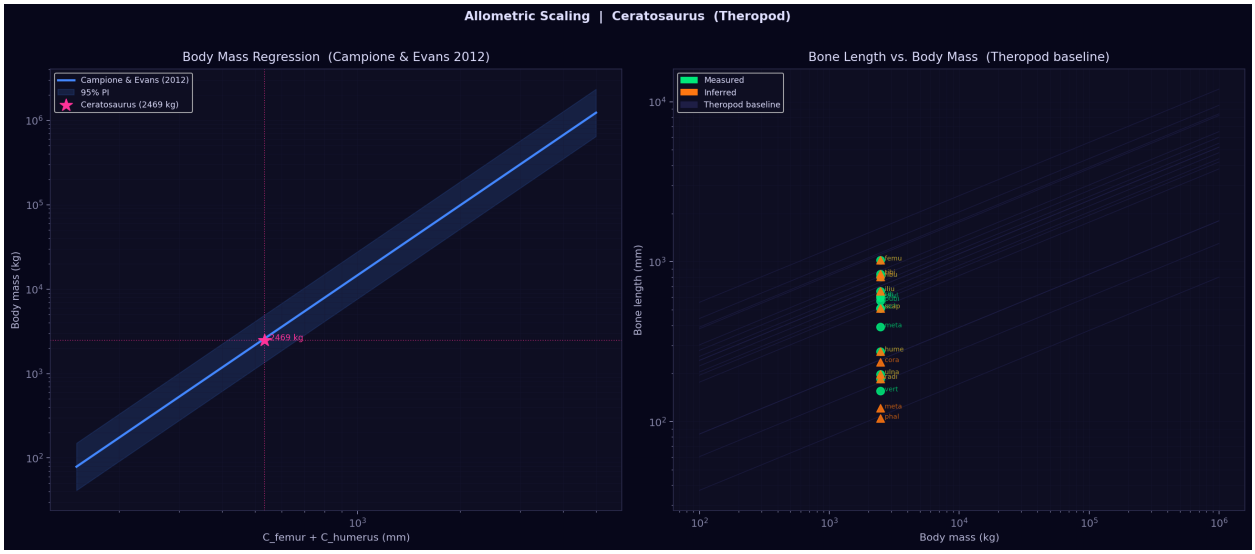



## Per-Element Stress Summary

| Element | σ_ax MPa | σ_bend MPa | σ_total MPa | SF | Status |
|---|---|---|---|---|---|
| coracoid | 0.030 | 0.056 | 0.085 | 15.000 | OK |
| femur | 2.171 | 25.195 | 27.367 | 6.212 | OK |
| fibula | 0.000 | 0.000 | 0.000 | 15.000 | OK |
| humerus | 0.895 | 6.350 | 7.245 | 15.000 | OK |
| ilium | 1.935 | 10.888 | 12.823 | 13.257 | OK |
| ischium | 1.465 | 9.437 | 10.902 | 15.000 | OK |
| metacarpal | 1.899 | 7.348 | 9.247 | 15.000 | OK |
| metatarsal | 9.995 | 30.443 | 40.438 | 4.204 | OK |
| phalanx | 0.000 | 0.000 | 0.000 | 15.000 | OK |
| pubis | 2.289 | 20.600 | 22.889 | 7.427 | OK |
| radius | 0.941 | 7.786 | 8.727 | 15.000 | OK |
| rib | 0.280 | 9.169 | 9.449 | 15.000 | OK |
| scapula | 0.022 | 0.075 | 0.096 | 15.000 | OK |
| skull | 0.000 | 0.000 | 0.000 | 15.000 | OK |
| tibia | 3.281 | 18.385 | 21.666 | 7.846 | OK |
| ulna | 0.827 | 6.866 | 7.693 | 15.000 | OK |
| vertebra | 2.171 | 3.829 | 6.000 | 15.000 | OK |

# 4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 ($R^2$=0.97; N=426 non-avian dinosaurs).



## 5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

| Parameter | Value | Notes |
|---|---|---|
| **Hip Height** | 2.245 m | Femur + tibia + metatarsal |
| **Estimated Speed** | 17.1 km/h  (4.76 m/s) | Gait: Trot \| Fr=1.029 |
| **Stride Length** | 5.194 m | Alexander (1976) / Thulborn (1990) |
| **Bite Force** | 2345.4 N  (239.1 kgf) | Calibrated theropod regression (Bates & Falkingham 2012) |
| **Skull Length** | 595.0 mm | Mid-shaft proxy from skull feature vector |
| **Femoral Robusticity** | 0.3634 | Circumference / length (dimensionless) |
| **Limb Strength Index** | 5.2e-05 | (A_f + A_t) / M^0.67  (Christiansen 1998) |
| **Est. Body Length** | 8.66 m | Allometric from femur length |
| **Est. Neck Length** | 1.21 m | Allometric from femur length |
| **Est. Tail Length** | 3.29 m | Allometric from femur length |

## 6 Complete Bone Manifest

| Element | L (mm) | D (mm) | Vol cm³ | Mass g | Method | Conf | SF |
|---|---|---|---|---|---|---|---|
| coracoid | 235.3 | 168.7 | 2682.84 | 3349.0 | Constraint | 0.40 | 15.00 |
| femur | 1020.0 | 118.0 | 5688.85 | 7101.4 | Measured | 0.88 | 6.21 |
| femur_right | 1020.0 | 118.0 | 5688.85 | 7101.4 | Symmetry | 0.84 | 6.21 |
| fibula | 805.0 | 40.0 | 364.17 | 381.9 | Measured | 0.88 | 15.00 |
| fibula_right | 805.0 | 40.0 | 364.17 | 381.9 | Symmetry | 0.84 | 15.00 |
| humerus | 275.0 | 52.0 | 297.85 | 371.8 | Measured | 0.88 | 15.00 |
| humerus_right | 275.0 | 52.0 | 297.85 | 371.8 | Symmetry | 0.84 | 15.00 |
| ilium | 655.0 | 125.0 | 4099.41 | 5117.3 | Measured | 0.88 | 13.26 |
| ilium_right | 655.0 | 125.0 | 4099.41 | 5117.3 | Symmetry | 0.84 | 13.26 |
| ischium | 510.0 | 85.0 | 1475.94 | 1842.4 | Measured | 0.88 | 15.00 |
| metacarpal | 121.2 | 34.6 | 21.64 | 17.8 | Constraint | 0.65 | 15.00 |
| metatarsal | 390.0 | 55.0 | 472.55 | 589.9 | Measured | 0.88 | 4.20 |
| phalanx | 104.6 | 35.1 | 15.59 | 12.1 | Constraint | 0.40 | 15.00 |
| pubis | 570.0 | 68.0 | 1055.73 | 1317.9 | Measured | 0.88 | 7.43 |
| radius | 185.0 | 30.0 | 66.69 | 83.3 | Measured | 0.88 | 15.00 |
| radius_right | 185.0 | 30.0 | 66.69 | 83.3 | Symmetry | 0.84 | 15.00 |
| rib | 610.0 | 30.0 | 219.90 | 274.5 | Measured | 0.88 | 15.00 |
| scapula | 510.0 | 198.0 | 8008.67 | 9997.2 | Measured | 0.88 | 15.00 |
| scapula_right | 510.0 | 198.0 | 8008.67 | 9997.2 | Symmetry | 0.84 | 15.00 |
| skull | 595.0 | 460.0 | 50430.43 | 62952.3 | Measured | 0.88 | 15.00 |
| tibia | 835.0 | 96.0 | 3082.40 | 3847.8 | Measured | 0.88 | 7.85 |
| tibia_right | 835.0 | 96.0 | 3082.40 | 3847.8 | Symmetry | 0.84 | 7.85 |
| ulna | 198.0 | 32.0 | 81.21 | 101.4 | Measured | 0.88 | 15.00 |
| ulna_right | 198.0 | 32.0 | 81.21 | 101.4 | Symmetry | 0.84 | 15.00 |
| vertebra | 155.0 | 118.0 | 864.48 | 1079.1 | Measured | 0.88 | 15.00 |